Welcome

NatEast Presents:

Fiducia

Trusted Insights. Confident Decisions.

A

Understanding of the problem statement

B

Simple credit risk formula used within your project

C

Solution design

D

The screen of the user interface

Understanding of the problem statement

Objective :

Design a simplified system that evaluates an individual's ability to repay a loan using financial and personal data.

How does Fiducia meet this:

- **AI Data Ingestion:** AI seamlessly captures and formats user inputs.
- **Inclusive Lending:** Built for fairness and accessibility for vulnerable users.
- **Deterministic Scoring:** Strict, transparent weighted-average formula eliminates bias.
- **Instant Pipeline:** Data saves to a DB, processes instantly, and outputs the score.

Required information

- ✓ Monthly salary
- ✓ Current savings
- ✓ Monthly mortgage payment
- ✓ Number of dependents
- ✓ Employment status
- ✓ Monthly credit card spending
- ✓ Other monthly loan repayments
- ✓ Dependents' ages
- ✓ Employment sector
- ✓ Job tenure (years)
- ✓ Housing status
- ✓ Savings trend
- ✓ Income variability
- ✓ Missed payments (last 12 months)
- ✓ Years of credit history
- ✓ Credit applications (last 6 months)

Credit Risk Report

Result: Low risk — 72.95 / 100

Per-factor breakdown

Category	Weight	Sub-score (0–100)	Weighted points	Basis
Affordability (DTI)	22.32%	15.8	3.53	Monthly debt 8,550 against salary 12,500: DTI = 68%.
Employment Stability	17.86%	86.25	15.4	Status: full time, 8 yr tenure, variable income.
Self-Declared Payment History	13.39%	100.0	13.39	0 self-declared missed payment(s) in the last 12 months.
Savings Buffer & Trend	13.39%	100.0	13.39	~6.8 months of cover; savings building up.
Self-Declared Credit History Length	8.04%	100.0	8.04	15 self-declared year(s) holding credit.
Housing Status	8.93%	75.0	6.7	Housing: own with mortgage.
Debt Composition (Credit Mix)	8.93%	80.0	7.14	3 active debt type(s).
Dependents Burden	4.46%	60.0	2.68	2 dependent(s), 2 under 18.
Self-Declared Recent Credit Activity	2.68%	100.0	2.68	0 self-declared credit application(s) in the last 6 months.
Total	**100.00%**		**72.95**	

What we Inferred from the problem statement:

What is a credit risk calculator:

A credit risk calculator estimates the likelihood or severity of a borrower failing to meet debt obligations.

Processes and factors which need to be taken into consideration:

- *The Whole Picture: A fair score which looks at your overall life such as your job stability and family commitments, not just your bank balance.*
- *Leftover Cash: It focuses on what you actually have left to spend each month after paying your mortgage, loans, and bills.*
- *Clear Rules: The calculation uses a straightforward formula so anyone can see exactly how the score was decided and also excludes biases which we took into consideration as well.*

Factors of consideration while building a credit risk calculator:

Key Inputs:

- Credit score, income, debt-to-income ratio
- Loan amount, term, collateral value

Key Outputs:

- Risk rating: Low, medium, High
- Probability of Default or Expected Loss

Implications:

- Few inputs, transparent logic, instant results
- Scorecard or weighted-formula approach
- Standalone tool - Web app

Inputs:

Why these inputs and how they matter:

- *Each input captures a distinct dimension of a borrower's ability and willingness to repay:*
- *Each input is defined by four components that work together to determine its influence on the final risk score*

How do these inputs affect credit score and how they give a better understand while selecting a credit score:

- ***Prioritises what matters most :*** *Lenders instantly are able to visually see which factors have a higher weighting which gives them clearer understanding of what to prioritize while producing a credit score.*
- ***Enables fair comparison:*** *Every applicant is scored on the same weighted scale.*
- ***Supports transparency:*** *Borrowers understand which factors to improve for a better score*

Category	Importance (1-5)	Reliability	Raw Score	Weight
Affordability (DTI)	5	1.0	5.0	22.32%
Employment Stability	4	1.0	4.0	17.86%
Self-Declared Payment History	5	0.6	3.0	13.39%
Savings Buffer & Trend	3	1.0	3.0	13.39%
Self-Declared Credit History Length	3	0.6	1.8	8.04%
Housing Status	2	1.0	2.0	8.93%
Debt Composition (Credit Mix)	2	1.0	2.0	8.93%
Dependents Burden	1	1.0	1.0	4.46%
Self-Declared Recent Credit Activity	1	0.6	0.6	2.68%

How is each factor weighted and taken into consideration during credit risk calculation:

Top Priorities for consideration :

- Affordability (22.32%): This checks your debt-to-income balance to ensure you have enough money left over to handle new payments.
- Employment Stability (17.86%): This allows lenders favor a reliable, long-term job history that promises a steady future income compared to lower firms or industries where income may not be stable which might mean that lenders will be more vary of consequences of lending

Middle Priorities:

- Payment History (13.39%): Tracking your past bill payments is highly important, but because this data is self-declared by the user rather than verified by a bank, its weight is slightly reduced.
- Savings Buffer & Trend (13.39%): This measures your emergency cash reserves. Having money saved up acts as a vital safety net if your income drops.
- Credit History Length (8.04%): This looks at how many years you have been managing credit. Like payment history, its impact is lowered because it relies on self-reported data.

Lower Priorities:

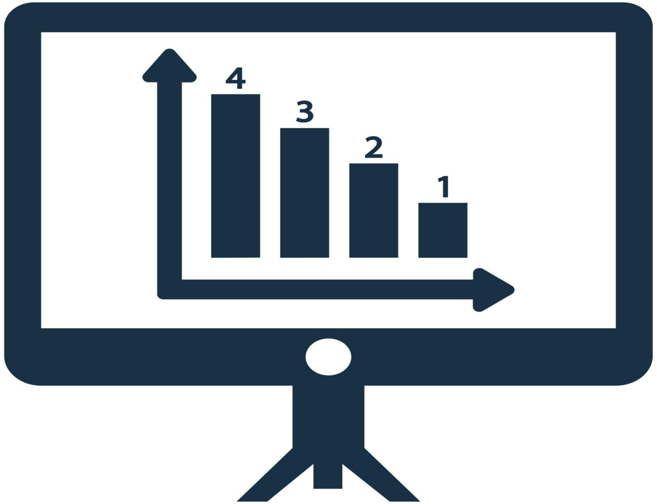
- Housing Status & Credit Mix (8.93% each): These look at whether you rent or own your home, and the variety of different accounts you manage. They help paint a fuller picture but don't move the score drastically.
- Dependents Burden (4.46%): This factor accounts for the baseline living costs of supporting family members or children.
- Recent Credit Activity (2.68%): This checks if you have applied for a lot of credit recently. It has the smallest overall impact on your final calculation.

Feature engineering

Context Matters: Raw financial figures don't tell the whole story; £1,000 of debt is high for someone earning £2,000, but minimal for someone earning £10,000.

Proportional Scaling: Converting raw data into ratios ensures the model evaluates an applicant's actual financial health, not just their surface numbers.

Fairer Evaluation: Using ratios makes the scoring system scalable, objective, and unbiased across all income levels.



Credit Risk Formula:

Key Benefits:

- Transparent :Clear, auditable logic showing how each factor contributes
- Fair : Every borrower evaluated on the same consistent scale
- Balanced : Considers strengths and weaknesses together, not just one metric
- Flexible : Weights can be adjusted as lending patterns evolve

```python
def calculate_risk(salary, savings, mortgage, loans, credit_spending,
    dependents, employment):

    # Avoid division errors
    if salary == 0:
        return "Invalid input"

    # Ratios
    dti = (mortgage + loans + credit_spending) / salary
    spending_ratio = credit_spending / salary
    savings_ratio = savings / salary

    # Employment risk score
    if employment.lower() == "full-time":
        emp_risk = 0
    elif employment.lower() == "part-time":
        emp_risk = 10
    else:
        emp_risk = 25

    # Risk score formula
    score = (0.4 * dti) + (0.2 * spending_ratio) + (0.1 * dependents) \
            - (0.2 * savings_ratio) + (0.1 * emp_risk)
```

What does the credit risk formula mean:

What the Code Does

This Python function calculates a credit risk score based on a borrower's financial inputs.

Validation: Returns "Invalid input" if salary is zero (prevents division errors)

2. Calculate Ratios

- `dti` = (mortgage + loans + credit_spending) / salary leads to Debt-to-Income
- `Spending ratio` = credit spending / salary leads to Spending burden
- `Savings ratio` = savings / salary leads to Financial buffer

3. Assign Employment Risk

Full-time : 0

Part-time: 10

Other: 25

4.Calculate final score:

score=(0.4×dti)+(0.2×spending)+(0.1×dependents)−(0.2×savings)+(0.1×emp_risk)

Input	Purpose
salary	Base income for calculating ratios
savings	Financial cushion available
mortgage	Monthly mortgage payment
loans	Existing loan obligations
Credit spending	Credit card or discretionary spending
dependents	Number of dependents
employment	Employment status (full-time, part-time, other)

Credit Risk Classification and what does the calculations mean ?

- **811 – 1,000 (Excellent):** Lowest risk. You get instant approvals, the highest credit limits, and the absolute lowest interest rates.
- **671 – 810 (Very Good):** Highly reliable. You easily qualify for most standard loans, cards, and leases with competitive rates.
- **531 – 670 (Good):** Acceptable history. You will get approved for most standard credit, but at average interest rates rather than top discounts.
- **439 – 530 (Fair):** Moderate risk. Getting approved is harder, requires closer scrutiny/paperwork, and comes with higher interest rates.
- **0 – 438 (Poor):** High risk. Standard approvals are very difficult to get; you must actively rebuild your score to secure credit.

How we eliminated Bias from the calculator:

1. Excluded Protected Characteristics and stereotypes:

The model uses only financial and behavioural inputs from the user:

- Age
- Gender
- Race or ethnicity
- Religion
- Marital status
- Location or postcode

However factors such as Age can be considered as biased since Younger borrowers might be penalised for shorter credit history however older borrowers penalised for retirement risk neither reflects actual repayment ability.Additionally, Gender has No legitimate link to credit worthiness which has been seen as a factor in the past.

What We Use Instead and why they prevented bias:

Measures actual ability to repay: Income, DTI, and payment history directly show whether someone can handle debt, regardless of who they are.

Avoids unfair assumptions : Age and gender are proxies for stereotypes, not facts. For example, A 22 year-old with stable income is no riskier than a 45 year-old with the same profile.

:Financial inputs measure behaviour and capacity; demographics measure assumptions. The calculator core focus is on what actually predicts repayment.rather than having biased viewpoints.

Features of the UI of Fiducia